

Electronic Devices and Circuits

1. Active Component – Device that requires power to operate (e.g., transistor).
2. Avalanche Breakdown – Sudden increase in current due to high reverse voltage.
3. Barrier Potential – Voltage required to overcome PN junction barrier.
4. Base – Middle region of a BJT controlling current flow.
5. Breakdown Voltage – Voltage at which diode conducts heavily in reverse.
6. Capacitor – Stores energy in electric field.
7. Collector – Region of BJT that collects charge carriers.
8. Cutoff Region – Transistor is OFF (no conduction).
9. Depletion Region – Area with no free charge carriers in PN junction.
10. Diode Current Equation – Relation between current and voltage in diode.
11. Drift Current – Current due to electric field.
12. Dynamic Resistance – Small-signal resistance of device.
13. Emitter – Region that emits charge carriers in BJT.
14. Energy Band Diagram – Shows energy levels of electrons.
15. Fermi Level – Energy level with 50% probability of occupancy.
16. Forward Resistance – Resistance during forward bias.
17. Hall Effect – Voltage generated due to magnetic field.
18. Hybrid Parameters (h-parameters) – Model for transistor analysis.
19. Intrinsic Semiconductor – Pure semiconductor material.
20. Junction Capacitance – Capacitance of PN junction.
21. Load Line – Graphical representation of circuit constraints.
22. Majority Carriers – Dominant charge carriers in material.
23. Minority Carriers – Less abundant charge carriers.
24. Ohmic Contact – Contact with linear V-I characteristics.
25. Passive Component – Device that does not require external power.
26. Reverse Saturation Current – Small current in reverse bias.
27. Small Signal Model – Linear approximation of device behavior.
28. Thermal Runaway – Increase in temperature causing higher current.
29. Transconductance – Ratio of output current to input voltage.
30. Zener Breakdown – Breakdown due to strong electric field.

Communication Systems

1. Amplitude Modulation (AM) – Varies amplitude of carrier signal.
2. Analog Communication – Transmission using continuous signals.
3. Antenna – Device that transmits or receives electromagnetic waves.
4. Attenuation – Reduction in signal strength over distance.
5. Bandwidth – Range of frequencies a signal occupies.
6. Baseband Signal – Original signal before modulation.
7. Carrier Signal – High-frequency signal used for transmission.
8. Channel – Medium through which signal travels.
9. Coherent Detection – Demodulation using synchronized carrier.
10. Demodulation – Process of retrieving original signal.
11. Digital Communication – Transmission using discrete signals.
12. Distortion – Alteration of signal shape.
13. Duplex Communication – Two-way communication system.
14. Encoding – Converting data into suitable format.
15. Frequency Modulation (FM) – Varies frequency of carrier signal.
16. Half Duplex – Two-way but one direction at a time.
17. Line Coding – Representation of digital data as signals.
18. Modulation – Process of combining signal with carrier.
19. Multiplexing – Combining multiple signals into one channel.
20. Noise – Unwanted disturbance in signal.
21. Phase Modulation (PM) – Varies phase of carrier signal.
22. Pulse Modulation – Modulation using pulse signals.
23. Sampling – Converting analog signal to discrete form.
24. Signal-to-Noise Ratio (SNR) – Ratio of signal power to noise power.
25. Simplex Communication – One-way communication.
26. Source – Origin of information.
27. Transmitter – Converts signal for transmission.
28. Receiver – Recovers original signal from received signal.
29. Transmission Line – Medium carrying electrical signals.
30. Wavelength – Distance between successive wave peaks.

Electromagnetic Field Theory

1. Ampere's Law – Relates magnetic field to current.
2. Boundary Conditions – Conditions at interface of two media.
3. Capacitance – Ability to store electric charge.
4. Charge Density – Charge per unit volume/area/length.
5. Conductivity – Measure of material's ability to conduct current.
6. Coulomb's Law – Force between two electric charges.
7. Current Density – Current per unit area.
8. Dielectric Constant – Ability of material to store electric energy.
9. Divergence – Measure of field spreading from a point.
10. Electric Field (E) – Force per unit charge.
11. Electric Flux – Flow of electric field through a surface.
12. Electric Potential – Work done per unit charge.
13. Faraday's Law – Changing magnetic field induces EMF.
14. Gauss's Law – Relates electric flux to enclosed charge.
15. Gradient – Rate of change of scalar field.
16. Inductance – Ability to store magnetic energy.
17. Lorentz Force – Force on charge in EM field.
18. Magnetic Field (H) – Field due to current or magnet.
19. Magnetic Flux – Total magnetic field through a surface.
20. Maxwell's Equations – Fundamental EM field equations.
21. Permeability – Ability to support magnetic field.
22. Permittivity – Ability to support electric field.
23. Poynting Vector – Represents power flow in EM wave.
24. Scalar Potential – Potential related to electric field.
25. Skin Effect – Current flows near conductor surface at high freq.
26. TEM Wave – Transverse electric and magnetic fields.
27. Transmission Line – Carries EM signals from one point to another.
28. Vector Potential – Helps in calculating magnetic field.
29. Wave Equation – Describes propagation of EM waves.
30. Wave Impedance – Ratio of electric to magnetic field.

Microprocessors

1. Accumulator – Register used for arithmetic and logic operations.
2. Address Bus – Carries address signals from CPU.
3. ALU (Arithmetic Logic Unit) – Performs arithmetic and logical operations.
4. Assembler – Converts assembly code into machine code.
5. Bus – Communication pathway for data transfer.
6. Cache Memory – High-speed memory for quick data access.
7. Clock – Synchronizes operations of the processor.
8. Control Unit – Directs execution of instructions.
9. Data Bus – Transfers data between components.
10. DMA (Direct Memory Access) – Transfers data without CPU.
11. Flag Register – Indicates status of operations.
12. Instruction – Command given to processor.
13. Instruction Cycle – Fetch, decode, execute sequence.
14. Instruction Set Architecture (ISA) – Set of instructions supported.
15. Interrupt – Signal that temporarily halts execution.
16. Interrupt Vector – Address of interrupt service routine.
17. I/O Ports – Interface for input and output devices.
18. Machine Language – Binary code executed by processor.
19. Memory – Stores data and instructions.
20. Microprocessor – CPU on a single integrated chip.
21. Opcode – Operation part of an instruction.
22. Operand – Data on which operation is performed.
23. Pipeline – Overlapping execution of instructions.
24. Program Counter (PC) – Holds address of next instruction.
25. Register – Small fast storage in CPU.
26. Stack – Memory structure for function calls.
27. Stack Pointer (SP) – Points to top of stack.
28. Throughput – Number of instructions processed per unit time.
29. Timing Diagram – Shows signal timing relationships.
30. Word Length – Number of bits processed at once.